

MSA - Most Common IATF Audit NC Findings

What auditors find and how to prevent every finding

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

Major NC Findings - MSA

- NC: GRR study not performed before first production run (Clause 7.1.5.1)
- NC: GRR result > 30% - gauge used for CC characteristic (Clause 8.5.1.1)
- NC: Gauge found in use with expired calibration label (Clause 7.1.5.2)
- NC: No MSA studies in PPAP submission package Element 8 (Clause 8.3.4.4)
- NC: GRR study only conducted with 1 operator - AV cannot be calculated (7.1.5.1)

Minor NC Findings - MSA

- NC: NDC not calculated or reported in GRR study
- NC: Bias study not performed for reference gauges
- NC: Linearity study not performed for gauges used across wide measurement range
- NC: No stability monitoring for critical gauges after calibration
- NC: Attribute GRR not performed for go/no-go gauges on CC characteristics
- NC: GRR study not repeated after gauge repair or 4M change
- NC: Gauge register not maintained - missing last/next calibration dates for some devices

How to Avoid MSA NCs

- Step 1: Maintain a complete Gauge Register with calibration and GRR status for every device.
- Step 2: Define MSA Plan in APQP Phase 3 - before any parts are produced.
- Step 3: Complete GRR, Bias, and Linearity before PPAP submission (Element 8).
- Step 4: Set up Stability monitoring for all critical gauges immediately after PPAP.
- Step 5: Trigger GRR re-study for any 4M change, gauge repair, or complaint linked to measurement.
- Step 6: Cross-reference: every CC characteristic must have GRR < 10% - verify the link.
- Step 7: Train all operators on measurement technique before GRR study - not after.